COMP3670/6670
Introduction to Machine Learning
Week 9: Eigen-decompositions & Principal Component Analysis

Yun Kuen Cheung, Marco
Senior Lecturer, School of Computing, ANU

Goals of these lectures

- Understand why we reduce the dimension of a dataset.
- Use eigenvectors to describe the variation captured by a covariance matrix.
- Find directions that retain the most variance.
- Encode and reconstruct data using PCA.
- Choose how many components to retain and interpret a 2D or 3D PCA plot.

Why reduce the number of features?

- $\mathbf{X} \in \mathbb{R}^{n \times d}$ stores n datapoints, each with d features.
- n counts samples; d is the ambient dimension.
- An MNIST image has $d = 784$ pixels, but nearby pixels often vary together.
- In footballer-performance dataset, features about defending (e.g., tackles, interceptions) are positively correlated.
- More features can help, but increase computation and make the space harder to cover.

Example 1. Suppose each of d features takes either 0 or 1, with all combinations allowed.

There are 2^d possible feature vectors.

$$2^{20} \;=\; 1{,}048{,}576.$$

With $n = 1000$ samples, we can observe at most 1000 distinct combinations: less than 0.1% of the possibilities.

- The curse of dimensionality: covering an unstructured feature space can require exponentially many samples.
- This does *not* mean every learning problem needs 2^d samples.
- Data may lie near a much smaller subspace.
- We seek $k < d$ coordinates that retain important variation.

Example 2 (cont'd). Suppose the useful signal lies on the line spanned by the unit vector $\mathbf{q}$.

$$\mathbf{q} = \frac{1}{5}\begin{pmatrix}3\\4\\0\end{pmatrix}, \qquad \mathbf{s} = \begin{pmatrix}15\\20\\0\end{pmatrix}, \qquad \mathbf{e} = \begin{pmatrix}0\\0\\3\end{pmatrix}.$$

We observe $\mathbf{x} = \mathbf{s} + \mathbf{e} = (15, 20, 3)^{\mathsf{T}}$.

The error $\mathbf{e}$ is perpendicular to the signal line, so projection recovers $\mathbf{s}$:

$$\langle \mathbf{q}, \mathbf{x} \rangle \, \mathbf{q} = 25\mathbf{q} = \mathbf{s}.$$

- Measurement errors and unwanted fluctuations are forms of noise.
- Projection removes components perpendicular to the retained subspace.
- PCA learns a subspace from observed variation, not from knowledge of the clean signal.
- Discarded variation is not automatically noise.

Eigenvectors of a square matrix

A matrix generally mixes an input vector's coordinates. We look for directions that it only scales.

Eigenvectors and eigenvalues. For $\mathbf{A} \in \mathbb{R}^{d \times d}$, a nonzero vector $\mathbf{v} \in \mathbb{R}^d$ is an eigenvector with eigenvalue $\lambda \in \mathbb{R}$ if

$$\mathbf{A}\mathbf{v} \;=\; \lambda \mathbf{v}.$$

If enough such directions form a basis, each coordinate is scaled separately.

- $\lambda > 0$: scale without reversing direction.
- $\lambda < 0$: scale and reverse direction.
- $\lambda = 0$: map the eigenvector to zero.
- The zero vector is not an eigenvector; otherwise every scalar would be an eigenvalue.

Eigenspace. For an eigenvalue λ,

$$\mathcal{E}_\lambda \;=\; \{\mathbf{v} \in \mathbb{R}^d : \mathbf{A}\mathbf{v} = \lambda\mathbf{v}\} \;=\; \mathrm{Nul}(\mathbf{A} - \lambda\mathbf{I}_d).$$

It is a subspace and includes the zero vector.

Every nonzero vector in it is an eigenvector.

Once λ is known, use Gaussian elimination on

$$(\mathbf{A} - \lambda\mathbf{I}_d)\mathbf{v} \;=\; \mathbf{0}.$$

Example 3 (cont'd). Consider

$$\mathbf{A} = \begin{pmatrix} 1 & 10 & -3 \\ 10 & 4 & 18 \\ -3 & 18 & -15 \end{pmatrix},$$

and the three column vectors

$$\mathbf{v}_1 = \begin{pmatrix} -1 \\ 2 \\ -3 \end{pmatrix}, \qquad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \qquad \mathbf{v}_3 = \begin{pmatrix} -4 \\ 1 \\ 2 \end{pmatrix}.$$

Check whether each vector is an eigenvector.

Example 3 (cont'd). The matrix-vector products are

$$\mathbf{A}\mathbf{v}_1 = \begin{pmatrix} 28 \\ -56 \\ 84 \end{pmatrix} = -28\mathbf{v}_1,$$

$$\mathbf{A}\mathbf{v}_2 = \begin{pmatrix} 18 \\ 36 \\ 18 \end{pmatrix} = 18\mathbf{v}_2, \qquad \mathbf{A}\mathbf{v}_3 = \mathbf{0}.$$

Their eigenvalues are -28, 18 and 0.

For large matrices, numerical software computes eigenvalues and eigenvectors.

Symmetric matrices have orthonormal eigenbases

Symmetric matrix. A real square matrix $\mathbf{A}$ is symmetric if

$$\mathbf{A}^\mathsf{T} = \mathbf{A},$$

or equivalently, $a_{ij} = a_{ji}$ for all i, j.

Not every real square matrix has a basis of real eigenvectors.

Symmetric matrices have an orthonormal eigenbasis, which preserves lengths and angles when we change coordinates.

Proposition 1. Eigenvectors of a real symmetric matrix corresponding to distinct eigenvalues are orthogonal.

Proof. Suppose

$$\mathbf{A}\mathbf{u} = \lambda\mathbf{u}, \qquad \mathbf{A}\mathbf{v} = \mu\mathbf{v}, \qquad \lambda \neq \mu.$$

Then...

Theorem 1: Spectral theorem. Every real symmetric $\mathbf{A} \in \mathbb{R}^{d \times d}$ has an orthonormal eigenbasis $\mathbf{q}_1, \ldots, \mathbf{q}_d$ with real eigenvalues $\lambda_1, \ldots, \lambda_d$.

Define

$$\mathbf{Q} \;=\; \begin{pmatrix} | & & | \\ \mathbf{q}_1 & \cdots & \mathbf{q}_d \\ | & & | \end{pmatrix}, \qquad \mathbf{\Lambda} \;=\; \operatorname{diag}(\lambda_1, \ldots, \lambda_d).$$

Then $\mathbf{Q}$ is orthogonal and

$$\mathbf{A} \;=\; \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\mathsf{T} \;=\; \sum_{j=1}^{d} \lambda_j \mathbf{q}_j \mathbf{q}_j^\mathsf{T}.$$

Example 4. Normalize the eigenvectors from Example 3, whose lengths are $\sqrt{14}$, $\sqrt{6}$ and $\sqrt{21}$.

Recall $\mathbf{A}$, and Stack the resulting unit eigenvectors to form $\mathbf{Q}$:

$$\mathbf{A} = \begin{pmatrix} 1 & 10 & -3 \\ 10 & 4 & 18 \\ -3 & 18 & -15 \end{pmatrix} \qquad \mathbf{Q} = \begin{pmatrix} -\frac{1}{\sqrt{14}} & \frac{1}{\sqrt{6}} & -\frac{4}{\sqrt{21}} \\ \frac{2}{\sqrt{14}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{21}} \\ -\frac{3}{\sqrt{14}} & \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{21}} \end{pmatrix}.$$

Then

$$\mathbf{A} = \mathbf{Q} \begin{pmatrix} -28 & 0 & 0 \\ 0 & 18 & 0 \\ 0 & 0 & 0 \end{pmatrix} \mathbf{Q}^{\mathsf{T}}.$$

Example 4 (cont'd).

$$\mathbf{A} = \mathbf{Q} \begin{pmatrix} -28 & 0 & 0 \\ 0 & 18 & 0 \\ 0 & 0 & 0 \end{pmatrix} \mathbf{Q}^{\mathsf{T}}.$$

If $\mathbf{x} = \alpha_1 \mathbf{q}_1 + \alpha_2 \mathbf{q}_2 + \alpha_3 \mathbf{q}_3$, then

$$\mathbf{A}\mathbf{x} = -28\alpha_1 \mathbf{q}_1 + 18\alpha_2 \mathbf{q}_2.$$

Positive semidefinite matrices. For a symmetric matrix $\mathbf{A}$, the scalar $\mathbf{v}^\mathsf{T}\mathbf{A}\mathbf{v}$ is its quadratic form at $\mathbf{v}$.

The matrix is positive semidefinite if

$$\mathbf{v}^\mathsf{T}\mathbf{A}\mathbf{v} \geq 0 \qquad \text{for every } \mathbf{v} \in \mathbb{R}^d.$$

This is the property we need for a matrix describing averages of squares.

Proposition 2. A real symmetric matrix is positive semidefinite exactly when all its eigenvalues are nonnegative.

Proof. ($\Leftarrow$) By the spectral theorem, write $\mathbf{v} = \sum_j \alpha_j \mathbf{q}_j$ in an orthonormal eigenbasis. Then

$$\mathbf{v}^\mathsf{T}\mathbf{A}\mathbf{v} \;=\; \sum_{j=1}^{d} \lambda_j \alpha_j^2.$$

If every $\lambda_j \geq 0$, the sum is nonnegative.

Conversely, substitute $\mathbf{v} = \mathbf{q}_j$:

$$\mathbf{q}_j^\mathsf{T}\mathbf{A}\mathbf{q}_j \;=\; \lambda_j.$$

Covariance captures how features vary together

We want variation between datapoints, not their common displacement from the origin.

Centering. For $n \geq 2$ datapoints,

$$\boldsymbol{\mu} \;=\; \frac{1}{n}\sum_{i=1}^{n} \mathbf{x}_i, \qquad \widetilde{\mathbf{x}}_i \;=\; \mathbf{x}_i - \boldsymbol{\mu}.$$

The centered data matrix is

$$\mathbf{X}_c \;=\; \mathbf{X} - \mathbf{1}\boldsymbol{\mu}^{\mathsf{T}} \in \mathbb{R}^{n\times d}.$$

Its rows are $\widetilde{\mathbf{x}}_i^{\mathsf{T}}$ and its feature columns have mean zero.

The vector $\mathbf{1} \in \mathbb{R}^n$ has every entry equal to 1.

Empirical variance. For numbers $t_1, \ldots, t_n$ with mean $\bar{t}$,

$$\mathrm{Var}(t_1, \ldots, t_n) \;=\; \frac{1}{n} \sum_{i=1}^{n} (t_i - \bar{t})^2.$$

Variance measures spread around the mean.

For centered observations, $\bar{t} = 0$, so variance is the average of their squares.

Covariance matrix. For centered data $\mathbf{X}_c$,

$$\mathbf{C} \;=\; \frac{1}{n}\mathbf{X}_c^{\mathsf{T}}\mathbf{X}_c \in \mathbb{R}^{d\times d}.$$

Its entries are

$$C_{j\ell} \;=\; \frac{1}{n}\sum_{i=1}^{n}(x_{ij}-\mu_j)(x_{i\ell}-\mu_\ell).$$

C_{jj} is feature j's variance. Off-diagonal entries describe how two features vary together.

We use $1/n$. Using $1/(n-1)$ scales all eigenvalues equally, but leaves PC directions and explained variance ratios unchanged.

Proposition 3. Every covariance matrix is symmetric and positive semidefinite.

Proof. The transpose rule gives

$$\mathbf{C}^\mathsf{T} \;=\; \frac{1}{n}(\mathbf{X}_c^\mathsf{T}\mathbf{X}_c)^\mathsf{T} \;=\; \frac{1}{n}\mathbf{X}_c^\mathsf{T}\mathbf{X}_c \;=\; \mathbf{C}.$$

For every $\mathbf{v} \in \mathbb{R}^d$,

$$\mathbf{v}^\mathsf{T}\mathbf{C}\mathbf{v} \;=\; \frac{1}{n}\|\mathbf{X}_c\mathbf{v}\|_2^2 \;\geq\; 0.$$

Example 5. Consider four datapoints in $\mathbb{R}^2$, stored as rows.

$$\mathbf{X} = \begin{pmatrix} 16 & 24 \\ 14 & 26 \\ 4 & 16 \\ 6 & 14 \end{pmatrix}, \qquad \boldsymbol{\mu} = \begin{pmatrix} 10 \\ 20 \end{pmatrix}.$$

Subtract the mean from each datapoint:

$$\mathbf{X}_c = \begin{pmatrix} 6 & 4 \\ 4 & 6 \\ -6 & -4 \\ -4 & -6 \end{pmatrix}.$$

Example 5 (cont'd). The covariance matrix is

$$\mathbf{C} = \frac{1}{4}\begin{pmatrix} 6 & 4 & -6 & -4 \\ 4 & 6 & -4 & -6 \end{pmatrix}\begin{pmatrix} 6 & 4 \\ 4 & 6 \\ -6 & -4 \\ -4 & -6 \end{pmatrix} = \begin{pmatrix} 26 & 24 \\ 24 & 26 \end{pmatrix}.$$

Both features have variance 26.

The positive covariance 24 means they tend to be above their means together and below their means together.

The directions $(1,1)^\mathsf{T}$ and $(1,-1)^\mathsf{T}$ have eigenvalues 50 and 2.

Example 5 (cont'd). Now consider a dataset with the same mean:

$$\mathbf{X}' = \begin{pmatrix} 16 & 14 \\ 14 & 16 \\ 4 & 26 \\ 6 & 24 \end{pmatrix}, \qquad \mathbf{X}'_c = \begin{pmatrix} 6 & -6 \\ 4 & -4 \\ -6 & 6 \\ -4 & 4 \end{pmatrix}.$$

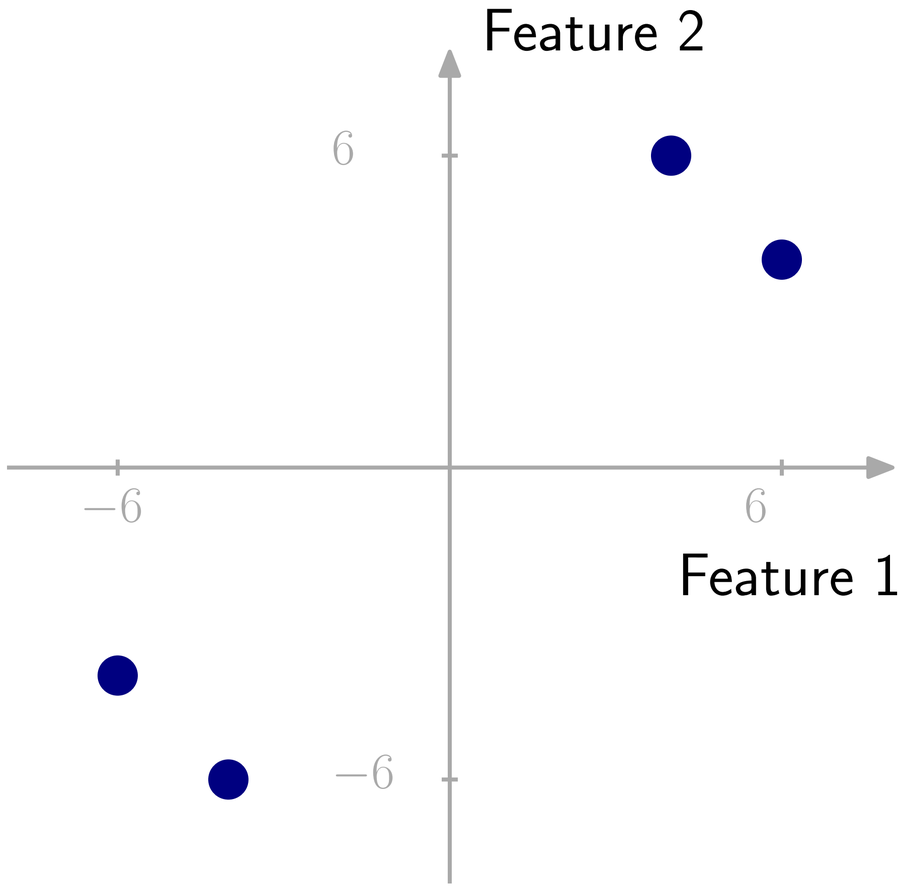
Its covariance matrix is

$$\mathbf{C}' = \frac{1}{4}(\mathbf{X}'_c)^\mathsf{T}\mathbf{X}'_c = \begin{pmatrix} 26 & -26 \\ -26 & 26 \end{pmatrix}.$$

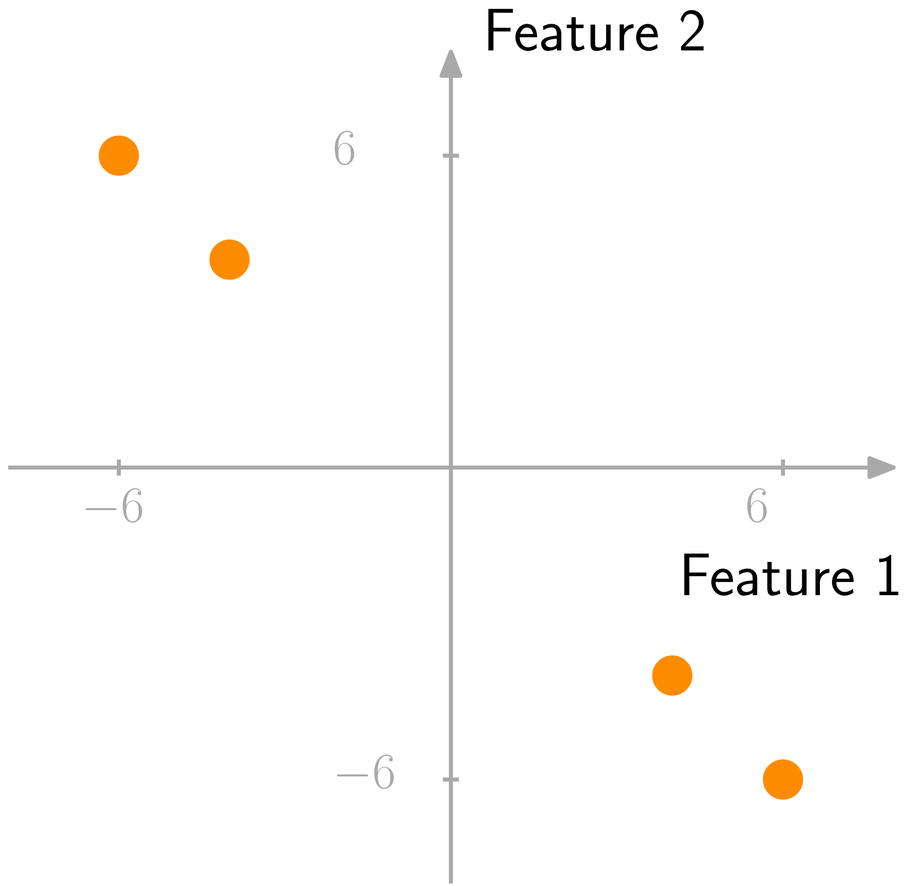
Here the features move in opposite directions and always sum to 30.

Example 5 (cont'd). Plot the centered datapoints.

Positive covariance: 24



Negative covariance: -26

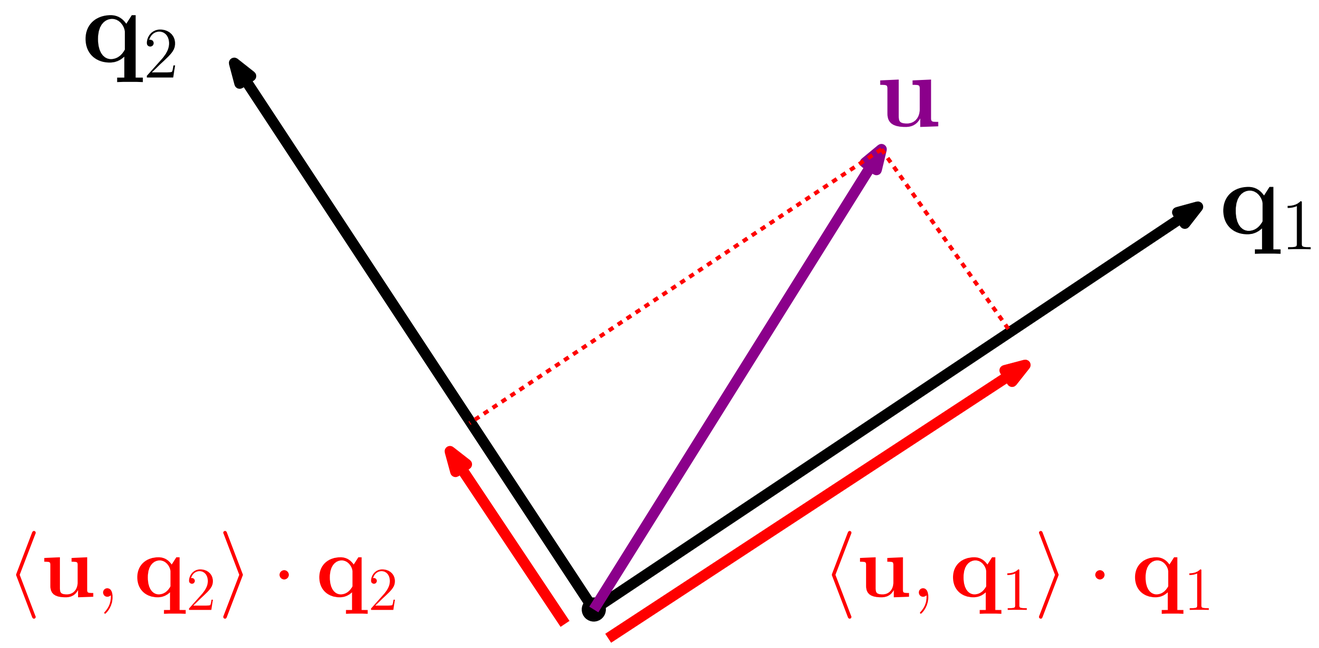


The first principal component retains the most variance

Suppose $\{\mathbf{q}_1, \mathbf{q}_2, \ldots, \mathbf{q}_d\}$ is an orthonormal basis of $\mathbb{R}^d$. Then any vector $\mathbf{u} \in \mathbb{R}^d$ can be uniquely expressed as

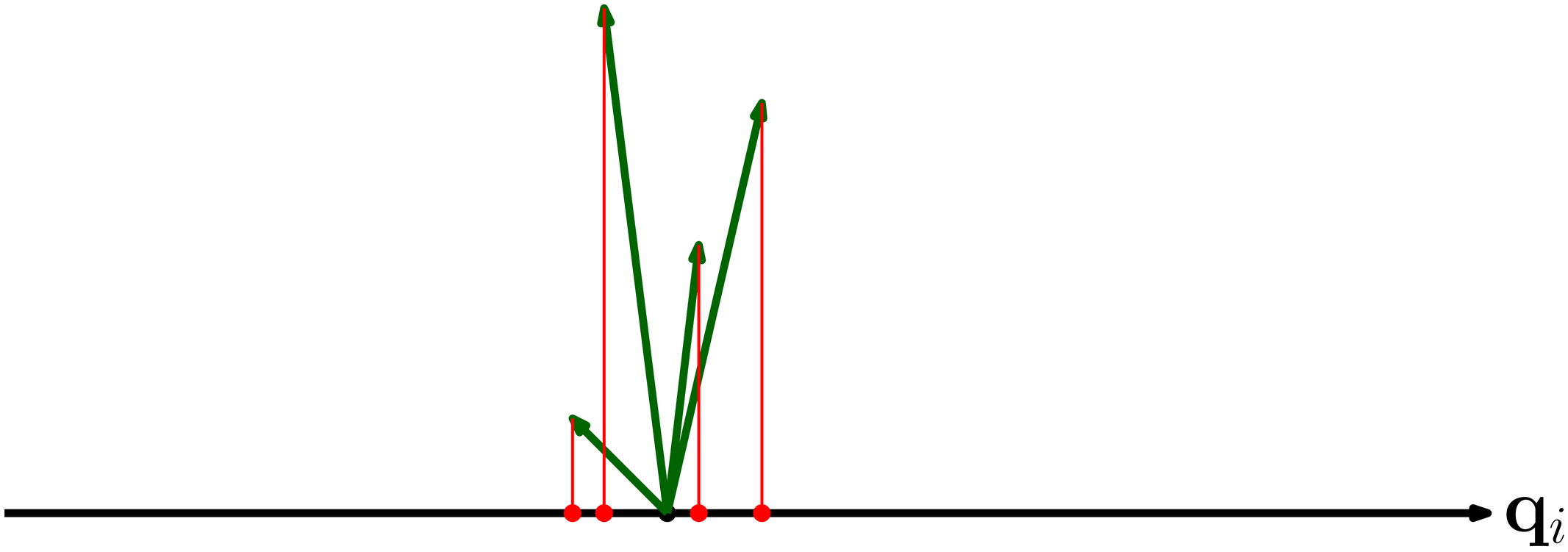
$$\mathbf{u} = c_1\mathbf{q}_1 + c_2\mathbf{q}_2 + \ldots + c_d\mathbf{q}_d,$$

where $c_i = \langle \mathbf{u}, \mathbf{q}_i \rangle$. We say $c_i\mathbf{q}_i$ is the projection of $\mathbf{u}$ along $\mathbf{q}_i$.

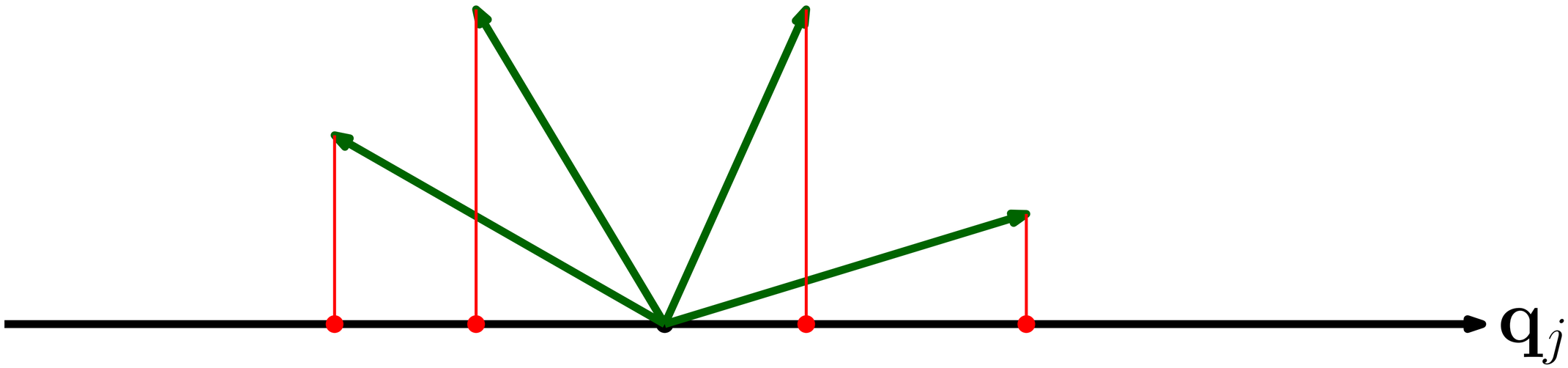


- When $d \geq 2$, there are infinitely many orthonormal bases.
- Informally speaking, dimensionality reduction is seeking an orthonormal basis, among which only the first few vectors are important for representing the data.
- What is meant by important? It is measured by variance of projection...

The red dots correspond to the projections of the 4 datapoints along $\mathbf{q}_i$. When the variance of the projections is small, they are typically close to each other and thus harder to distinguish. This indicates projecting along $\mathbf{q}_i$ is not good/important.



If we can find another vector $\mathbf{q}_j$ such that the projected red dots are far away from each other and thus easier to distinguish, projecting along $\mathbf{q}_j$ is good/important.



To store one number per centered datapoint, choose a unit direction $\mathbf{q}$ and keep $t_i = \langle \mathbf{q}, \widetilde{\mathbf{x}}_i \rangle$.

Proposition 4. These scores have mean zero and variance

$$\mathrm{Var}(t_1, \ldots, t_n) \;=\; \frac{1}{n} \sum_i \langle \mathbf{q}, \widetilde{\mathbf{x}}_i \rangle^2 \;=\; \mathbf{q}^\mathsf{T} \mathbf{C} \mathbf{q}.$$

Proof. The centered datapoints sum to zero, and the score vector is $\mathbf{t} = \mathbf{X}_c \mathbf{q}$.

Unit length matters: scaling $\mathbf{q}$ by a scales variance by a^2.

First principal component. A first principal-component direction is a unit vector maximizing $\mathbf{q}^\mathsf{T}\mathbf{C}\mathbf{q}$.

Theorem 2. Order an orthonormal covariance eigenbasis by

$$\lambda_1 \geq \cdots \geq \lambda_d \geq 0.$$

Then

$$\max_{\|\mathbf{q}\|_2=1} \mathbf{q}^\mathsf{T}\mathbf{C}\mathbf{q} = \lambda_1.$$

The maximizers are exactly the unit vectors in the eigenspace of λ_1. In particular, $\mathbf{q}_1$ is a first PC direction.

Proof of Theorem 2. Expand a unit vector in the orthonormal eigenbasis:

$$\mathbf{q} \;=\; \sum_{j=1}^{d} \alpha_j \mathbf{q}_j, \qquad \sum_{j=1}^{d} \alpha_j^2 \;=\; 1.$$

Therefore

$$\mathbf{q}^{\mathsf{T}}\mathbf{C}\mathbf{q} \;=\; \sum_{j=1}^{d} \lambda_j \alpha_j^2 \;\leq\; \lambda_1 \sum_{j=1}^{d} \alpha_j^2 \;=\; \lambda_1.$$

Equality holds exactly when $\alpha_j = 0$ whenever $\lambda_j < \lambda_1$.

Example 6. For the remaining PCA examples, use six datapoints in $\mathbb{R}^4$.

$$\mathbf{X} = \begin{pmatrix} 38 & 43 & 41 & 50 \\ 20 & 31 & 53 & 68 \\ 26 & 40 & 50 & 56 \\ 19 & 30 & 22 & 37 \\ 7 & 12 & 40 & 49 \\ 10 & 24 & 34 & 40 \end{pmatrix}, \quad \boldsymbol{\mu} = \begin{pmatrix} 20 \\ 30 \\ 40 \\ 50 \end{pmatrix}, \quad \mathbf{X}_c = \begin{pmatrix} 18 & 13 & 1 & 0 \\ 0 & 1 & 13 & 18 \\ 6 & 10 & 10 & 6 \\ -1 & 0 & -18 & -13 \\ -13 & -18 & 0 & -1 \\ -10 & -6 & -6 & -10 \end{pmatrix}.$$

Example 6 (cont'd). The covariance matrix is

$$\mathbf{C} = \frac{1}{6}\mathbf{X}_c^\top \mathbf{X}_c = \begin{pmatrix} 105 & 98 & 26 & 27 \\ 98 & 105 & 27 & 26 \\ 26 & 27 & 105 & 98 \\ 27 & 26 & 98 & 105 \end{pmatrix}.$$

Its ordered eigenvalues are

$$\lambda_1 = 256, \qquad \lambda_2 = 150, \qquad \lambda_3 = 8, \qquad \lambda_4 = 6.$$

Each original feature has variance 105.

Example 6 (cont'd). The first PC direction and its scores are

$$\mathbf{q}_1 \;=\; \frac{1}{2}\begin{pmatrix}1\\1\\1\\1\end{pmatrix}, \qquad \mathbf{t} \;=\; \mathbf{X}_c\mathbf{q}_1 \;=\; \begin{pmatrix}16\\16\\16\\-16\\-16\\-16\end{pmatrix}.$$

Their mean is zero and their variance is

$$\frac{1}{6}\sum_{i=1}^{6} t_i^2 \;=\; \frac{1}{6}\times(6\times 16^2) \;=\; 256.$$

This exceeds the variance 105 of any individual feature.

Recap: one principal component

- The largest covariance eigenvalue is the maximum variance along a unit direction.
- Changing $\mathbf{q}_1$ to $-\mathbf{q}_1$ reverses all scores, but leaves the line and reconstruction errors unchanged.
- If the largest eigenvalue repeats, several different directions may maximize variance.

Several principal components retain a subspace

Repeating the first direction would store the same information repeatedly. Instead, require mutually orthogonal directions.

PCA basis matrix. Order orthonormal covariance eigenvectors by decreasing eigenvalue and choose $1 \leq k \leq d$.

$$\mathbf{Q}_k = \begin{pmatrix} | & & | \\ \mathbf{q}_1 & \cdots & \mathbf{q}_k \\ | & & | \end{pmatrix} \in \mathbb{R}^{d \times k}.$$

The retained subspace is $\mathcal{S}_k = \operatorname{span}\{\mathbf{q}_1, \ldots, \mathbf{q}_k\}$.

PCA coordinates. A datapoint is encoded as

$$\mathbf{z}_i \;=\; \mathbf{Q}_k^{\mathsf{T}}(\mathbf{x}_i - \boldsymbol{\mu}) \in \mathbb{R}^k.$$

For the entire training batch,

$$\mathbf{Z} \;=\; \mathbf{X}_c\mathbf{Q}_k \in \mathbb{R}^{n\times k}.$$

The i-th row of $\mathbf{Z}$ is $\mathbf{z}_i^{\mathsf{T}}$.

These are coordinates of the projection, not necessarily an exact representation of the whole datapoint.

Proposition 5. The direction $\mathbf{q}_j$ maximizes variance among unit vectors orthogonal to $\mathbf{q}_1, \ldots, \mathbf{q}_{j-1}$. The maximum is λ_j.

The retained scores have mean zero and covariance

$$\frac{1}{n}\mathbf{Z}^\top\mathbf{Z} \;=\; \mathbf{Q}_k^\top \mathbf{C}\mathbf{Q}_k \;=\; \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & \cdots & \lambda_k \end{pmatrix}.$$

Each retained coordinate contributes variance λ_j.

Distinct score coordinates have zero covariance: they are uncorrelated.

Proof of Proposition 5. A unit vector orthogonal to the first $j-1$ eigenvectors has the expansion

$$\mathbf{q} \;=\; \sum_{\ell=j}^{d} \alpha_\ell \mathbf{q}_\ell, \qquad \sum_{\ell=j}^{d} \alpha_\ell^2 \;=\; 1.$$

Hence $\mathbf{q}^\mathsf{T}\mathbf{C}\mathbf{q} = \sum_{\ell=j}^{d} \lambda_\ell \alpha_\ell^2 \le \lambda_j$, with equality for $\mathbf{q} = \mathbf{q}_j$.

For the score covariance, use $\mathbf{Z} = \mathbf{X}_c \mathbf{Q}_k$ and

$$\mathbf{q}_j^\mathsf{T}\mathbf{C}\mathbf{q}_\ell \;=\; \lambda_\ell \langle \mathbf{q}_j, \mathbf{q}_\ell \rangle \;=\; \begin{cases} \lambda_j, & j = \ell, \\ 0, & j \neq \ell. \end{cases}$$

Example 7. For the six-point dataset, an ordered orthonormal eigenbasis is

$$\mathbf{q}_1 = \frac{1}{2}\begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \ \mathbf{q}_2 = \frac{1}{2}\begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix}, \ \mathbf{q}_3 = \frac{1}{2}\begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix}, \ \mathbf{q}_4 = \frac{1}{2}\begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix}.$$

The corresponding eigenvalues are 256, 150, 8 and 6.

Keep $\mathbf{q}_1$ and $\mathbf{q}_2$; omit the much smaller-variance directions $\mathbf{q}_3$ and $\mathbf{q}_4$.

Example 7 (cont'd). Keeping two directions gives

$$\mathbf{Q}_2 \;=\; \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 1 & -1 \\ 1 & -1 \end{pmatrix}, \qquad \mathbf{Z} \;=\; \mathbf{X}_c\mathbf{Q}_2 \;=\; \begin{pmatrix} 16 & 15 \\ 16 & -15 \\ 16 & 0 \\ -16 & 15 \\ -16 & -15 \\ -16 & 0 \end{pmatrix}.$$

For example, $\mathbf{x}_1$ is encoded as

$$\mathbf{z}_1 \;=\; \begin{pmatrix} 16 \\ 15 \end{pmatrix}.$$

The two score columns have variances 256 and 150.

PCA minimizes reconstruction error

Lecture 5 reconstructs a projection by multiplying its orthonormal coordinates by the basis matrix. Here we must also restore the mean.

Reconstruction. For $\mathbf{U} \in \mathbb{R}^{d \times k}$ with orthonormal columns,

$$\widehat{\mathbf{x}}_i \;=\; \boldsymbol{\mu} + \mathbf{U}\mathbf{U}^{\mathsf{T}}(\mathbf{x}_i - \boldsymbol{\mu}).$$

The average squared reconstruction error is

$$E(\mathbf{U}) \;=\; \frac{1}{n}\sum_{i=1}^{n} \|\mathbf{x}_i - \widehat{\mathbf{x}}_i\|_2^2.$$

Theorem 3: PCA optimality. Let $\lambda_1 \geq \cdots \geq \lambda_d \geq 0$ be the covariance eigenvalues and $1 \leq k \leq d$.

For any orthonormal choice $\mathbf{U} = \begin{pmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_k \end{pmatrix}$,

$$E(\mathbf{U}) \;\geq\; \sum_{\ell=k+1}^{d} \lambda_\ell.$$

Equality is attained by $\mathbf{U} = \mathbf{Q}_k$.

PCA both maximizes retained variance and minimizes reconstruction error.

Example 8. Reconstruct the first datapoint using the two scores from Example 7:

$$\widehat{\mathbf{x}}_1 \;=\; \boldsymbol{\mu} + 16\mathbf{q}_1 + 15\mathbf{q}_2 \;=\; \frac{1}{2}\begin{pmatrix} 71 \\ 91 \\ 81 \\ 101 \end{pmatrix}.$$

The residual is

$$\mathbf{x}_1 - \widehat{\mathbf{x}}_1 \;=\; \frac{1}{2}\begin{pmatrix} 5 \\ -5 \\ 1 \\ -1 \end{pmatrix}.$$

Example 8 (cont'd). The full reconstruction and residual matrices are

$$\widehat{\mathbf{X}} = \frac{1}{2}\begin{pmatrix} 71 & 91 & 81 & 101 \\ 41 & 61 & 111 & 131 \\ 56 & 76 & 96 & 116 \\ 39 & 59 & 49 & 69 \\ 9 & 29 & 79 & 99 \\ 24 & 44 & 64 & 84 \end{pmatrix}, \quad \mathbf{X}-\widehat{\mathbf{X}} = \frac{1}{2}\begin{pmatrix} 5 & -5 & 1 & -1 \\ -1 & 1 & -5 & 5 \\ -4 & 4 & 4 & -4 \\ -1 & 1 & -5 & 5 \\ 5 & -5 & 1 & -1 \\ -4 & 4 & 4 & -4 \end{pmatrix}.$$

Each row of the residual matrix gives one datapoint's reconstruction error vector.

Example 8 (cont'd). The six squared reconstruction errors are

$$13, \quad 13, \quad 16, \quad 13, \quad 13, \quad 16.$$

Their average is

$$E(\mathbf{Q}_2) \;=\; \frac{1}{6}(13+13+16+13+13+16) \;=\; 14 \;=\; \lambda_3+\lambda_4.$$

Keeping only the first PC would give average error

$$E(\mathbf{Q}_1) \;=\; 150+8+6 \;=\; 164.$$

Corollary: rank and exact reconstruction. If $\mathbf{X}_c$ has rank r, its covariance matrix has exactly r positive eigenvalues.

$$\operatorname{rank}(\mathbf{C}) \;=\; \operatorname{rank}(\mathbf{X}_c) \;=\; r \;\leq\; \min(d, n-1).$$

Retaining all r positive-eigenvalue directions reconstructs every training datapoint exactly.

If $r = 0$, the mean alone already reconstructs every datapoint.

- When $d \geq n$, centered training data always lie in a subspace of dimension at most $n - 1$.
- This holds even when the observations consist entirely of noise.
- Exact reconstruction using $n - 1$ PCs is not evidence of meaningful low-dimensional structure.
- Ask whether a much smaller representation retains useful structure and works for new datapoints.

Using PCA on a dataset

Explained variance ratio. When $\sum_{j=1}^{d} \lambda_j > 0$, the fraction of training variance retained by k PCs is

$$\rho_k = \frac{\sum_{j=1}^{k} \lambda_j}{\sum_{j=1}^{d} \lambda_j}.$$

The numerator is retained variance; the denominator is total centered variance.

One rule: choose the smallest k that reaches a selected threshold.

Example 9. For our six-point dataset, retain the first two PCs:

$$\rho_2 = \frac{256 + 150}{256 + 150 + 8 + 6} = \frac{406}{420} \approx 0.9667.$$

This retains about 96.67% of the training variance.

It meets a 95% threshold, but not a 99% threshold.

The discarded fraction is $1/30$, corresponding to average squared reconstruction error 14.

Algorithm 1: PCA with k retained components.

Require: Training data $\mathbf{X} \in \mathbb{R}^{n \times d}$ and $1 \leq k \leq d$

1: Compute $\boldsymbol{\mu} = \frac{1}{n} \sum_i \mathbf{x}_i$
2: Center: $\mathbf{X}_c = \mathbf{X} - \mathbf{1}\boldsymbol{\mu}^\mathsf{T}$
3: Optionally scale the centered features; retain the scaling parameters
4: Form $\mathbf{C} = \frac{1}{n}\mathbf{X}_c^\mathsf{T}\mathbf{X}_c$
5: Obtain orthonormal eigenvectors ordered by decreasing eigenvalue
6: Form $\mathbf{Q}_k$ from the first k eigenvectors
7: Compute $\mathbf{Z} = \mathbf{X}_c\mathbf{Q}_k$
8: Retain $\boldsymbol{\mu}$, $\mathbf{Q}_k$ and any scaling parameters

Numerical software finds the eigenvalues and eigenvectors.

Encoding and reconstruction. Without optional feature scaling, a new datapoint uses

$$\mathbf{z} = \mathbf{Q}_k^\top(\mathbf{x} - \boldsymbol{\mu}), \qquad \widehat{\mathbf{x}} = \boldsymbol{\mu} + \mathbf{Q}_k\mathbf{z}.$$

For the training batch,

$$\widehat{\mathbf{X}} = \mathbf{1}\boldsymbol{\mu}^\top + \mathbf{Z}\mathbf{Q}_k^\top.$$

Fit the mean and directions on training data only.

Use the same fitted transformation for validation, test and future datapoints.

- Feature units matter: changing metres to millimetres multiplies variance by 10^6.
- Optionally scale the centered features according to the application.
- Apply fitted scaling before encoding; undo it before adding the mean during reconstruction.
- After scaling, variance and reconstruction-error optimality refer to the scaled feature space.

What PCA can and cannot promise

- Fewer features can reduce storage and the cost of subsequent models.
- Discarding low-variance directions can reduce noise when those directions mainly contain noise.
- PCA does not use class labels.
- High variance need not be useful for prediction; a low-variance direction may distinguish classes.

Visualizing high-dimensional data

A high-dimensional datapoint cannot be displayed directly as a point in 2D or 3D.

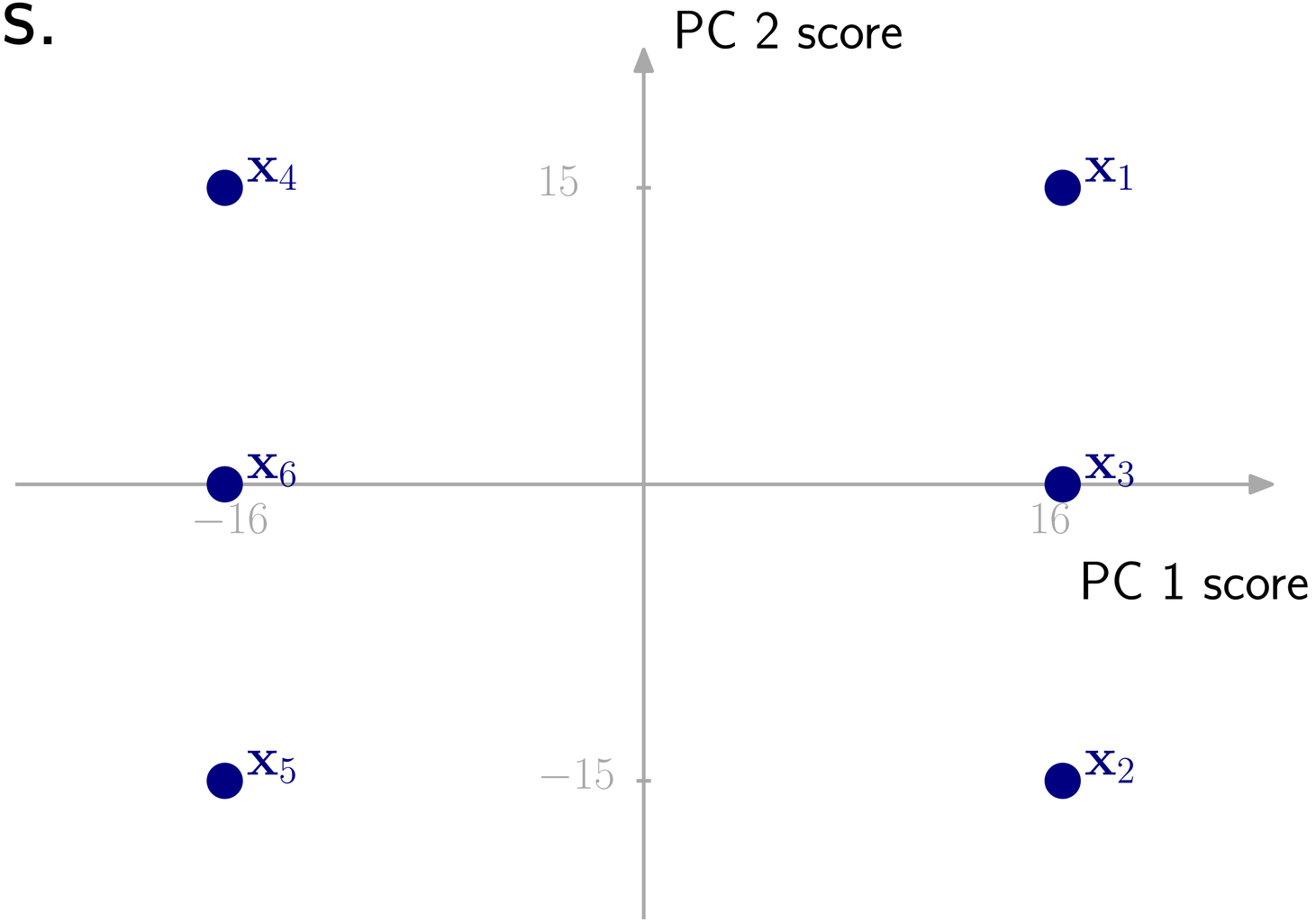
Plot its first two PCA scores:

$$\mathbf{z}_i = \begin{pmatrix} \langle \mathbf{q}_1, \mathbf{x}_i - \boldsymbol{\mu} \rangle \\ \langle \mathbf{q}_2, \mathbf{x}_i - \boldsymbol{\mu} \rangle \end{pmatrix}.$$

For a 3D plot, also retain the score along $\mathbf{q}_3$.

Each axis is a linear combination of features, not simply a selected original feature.

Example 10. Visualize the six datapoints in $\mathbb{R}^4$ using two PCs.



Two groups are separated along PC 1.

Variation within each group appears along PC 2.

These axes retain 96.67% of the training variance.

Variation along PC 3 and PC 4 is invisible.

Example 11.

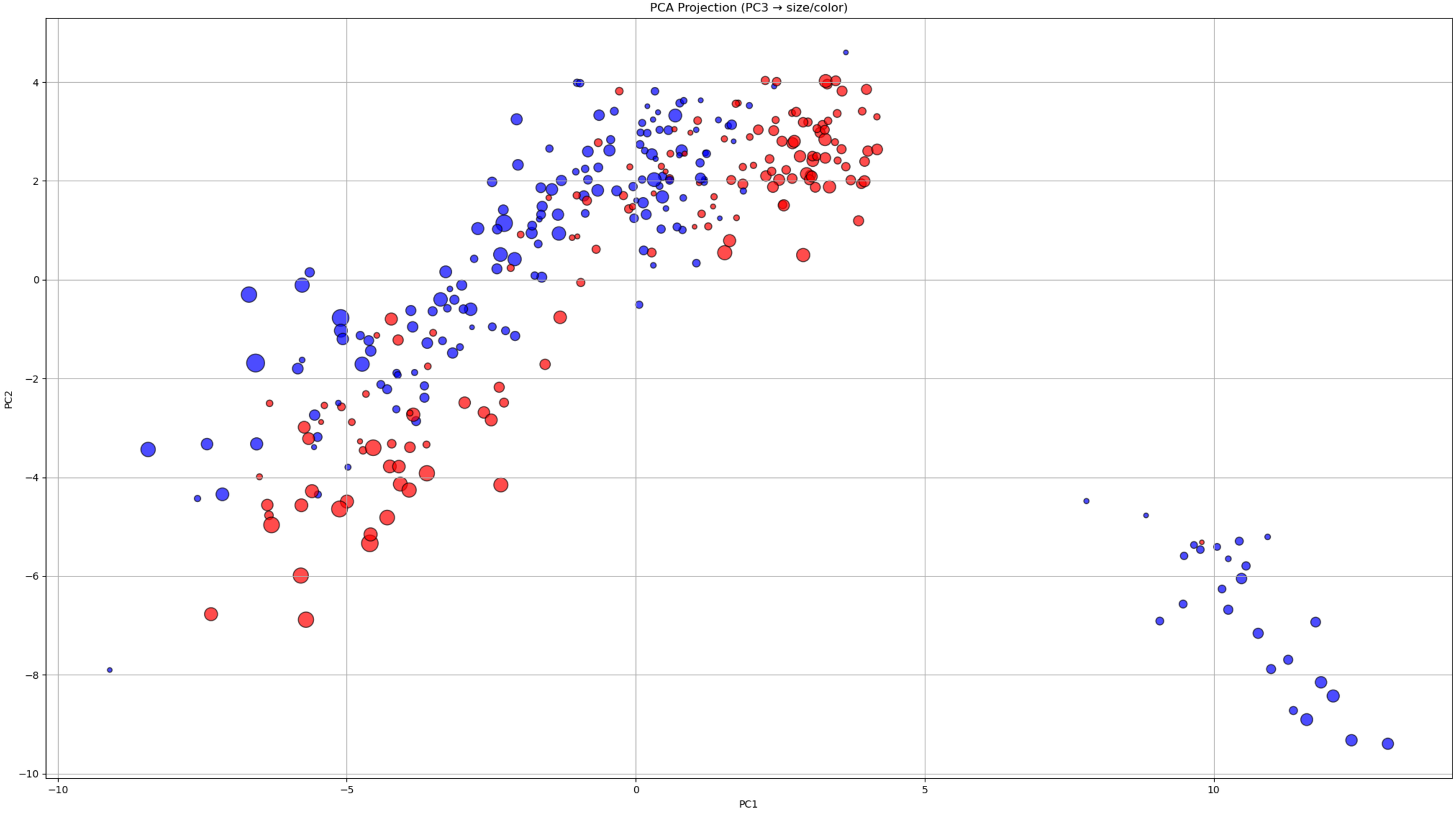
I have a dataset of players' performance stat for the English Premier League 2024/25. (The dataset is not freely available, so I can't share with you.)

For each player, there are 77 features, some examples:

- "minutesPlayed"
- "accurateLongBalls"
- "accurateLongBallsPercentage"
- "saves" (only goalkeeper has non-zero saves)

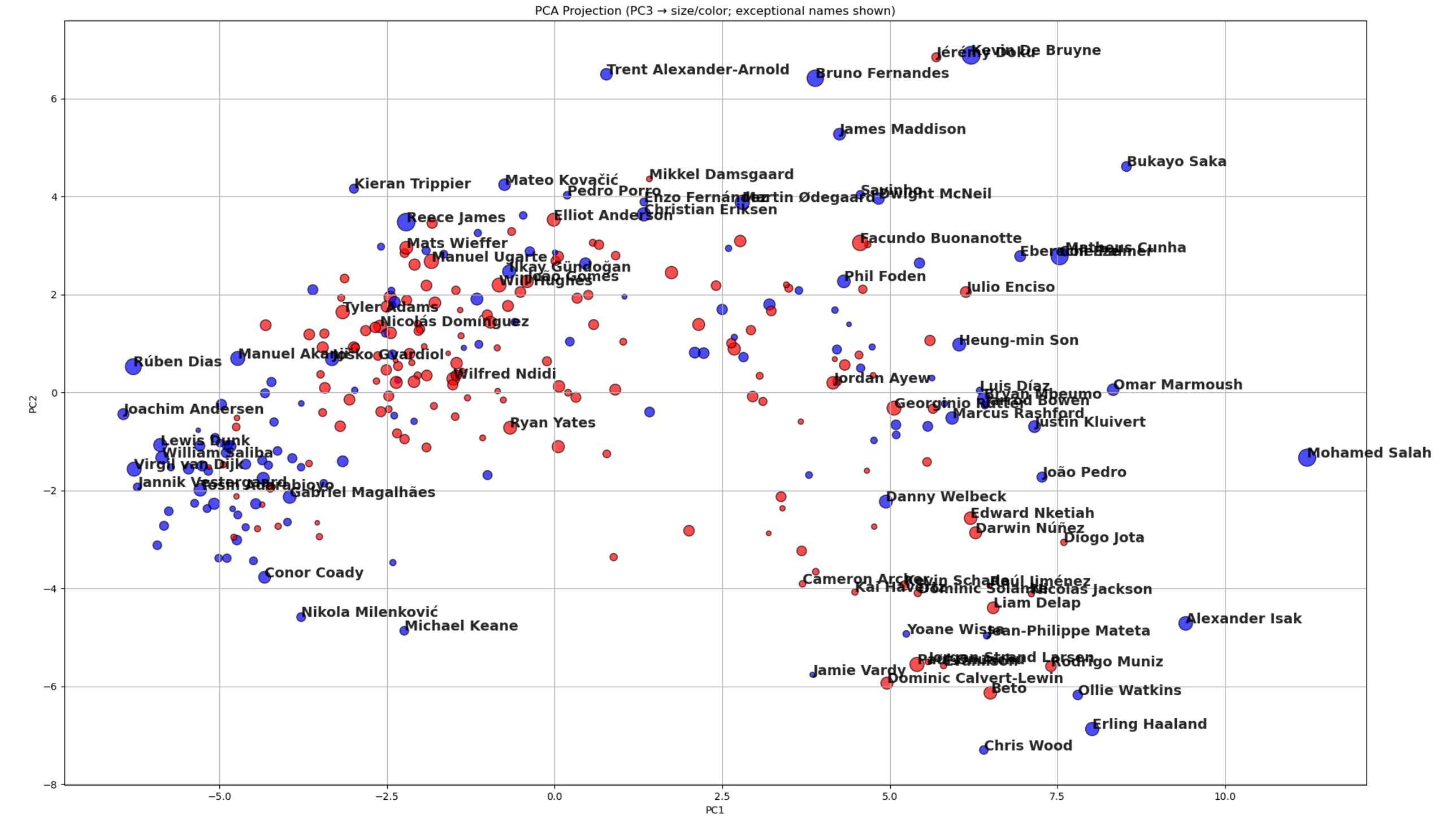
Example 11 (con't).

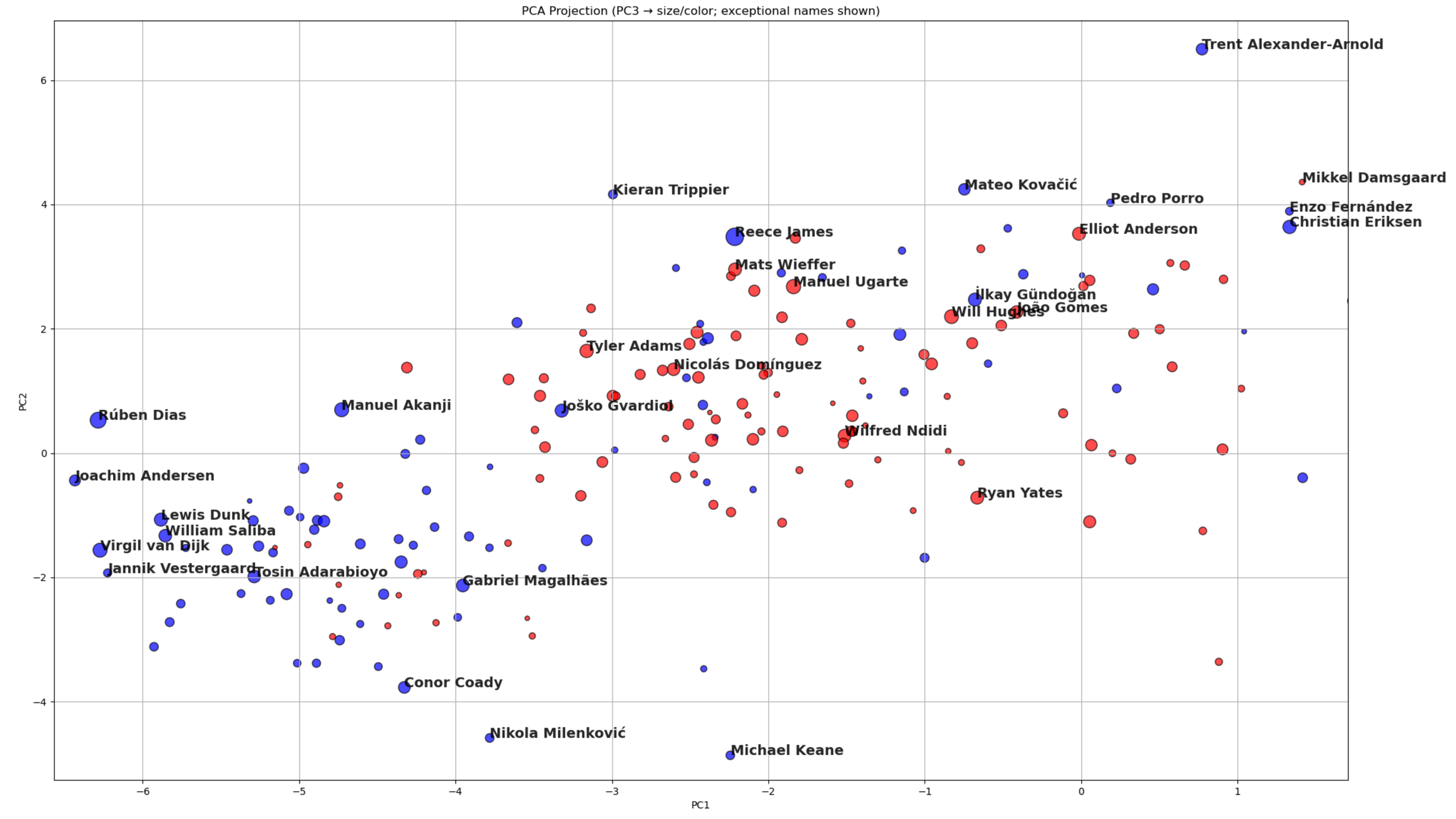
- I am only interested in players with substantial participation, so I remove all players who played < 900 minutes.
- Still, some player played just 900 minutes, while another player played more than 3000 minutes.
- To ensure fair comparison, for any non-percentage feature (like "accurateLongBalls"), divide it by "minutesPlayed".
- I remove features clearly irrelevant to distinguishing types of players, e.g., "appearances", "matchesStarted".

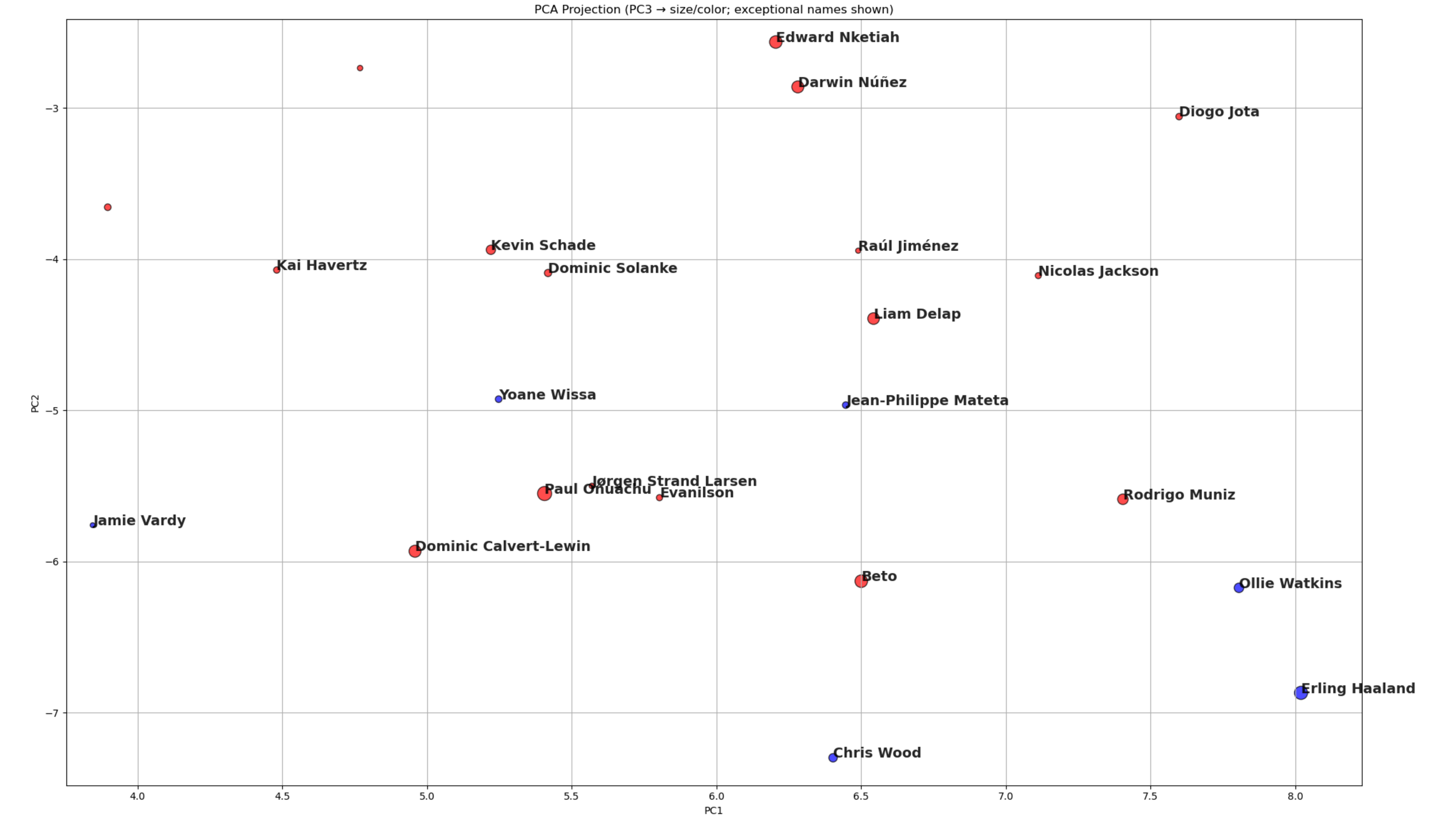

PCA Projection (PC3 → size/color)
PC1
PC2
−10
−5
0
5
10
4
2
0
−2
−4
−6
−8
−10

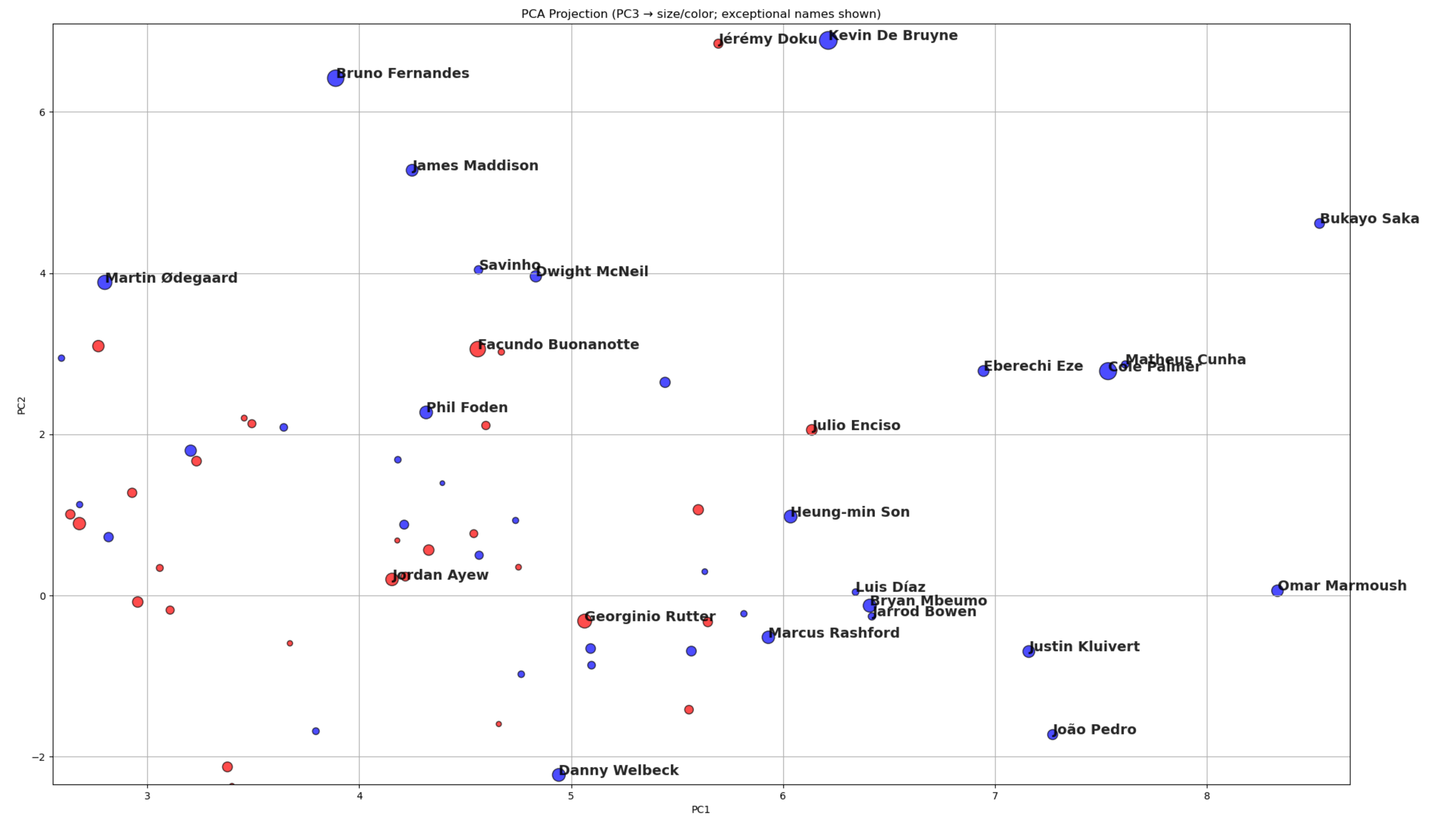
Example 11 (con't).

- So I remove all goalkeepers and goalkeeper-specific features, e.g., "saves", "punches".
- Exceptional players in at least one of the three dimensions are named in the plot.


PCA Projection (PC3 → size/color; exceptional names shown)
PC1
PC2
Kevin De Bruyne
Jérémy Doku
Trent Alexander-Arnold
Bruno Fernandes
James Maddison
Bukayo Saka
Mikkel Damsgaard
Kieran Trippier
Mateo Kovačić
Pedro Porro
Enzo Fernández
Martin Ødegaard
Saïnho
Dwight McNeil
Christian Eriksen
Reece James
Elliot Anderson
Facundo Buonanotte
Mats Wieffer
Manuel Ugarte
Matheus Cunha
Ilkay Gündoğan
João Gomes
Phil Foden
Julio Enciso
Tyler Adams
Nicolás Domínguez
Heung-min Son
Rúben Dias
Manuel Akanji
Wilfred Ndidi
Jordan Ayew
Omar Marmoush
Joachim Andersen
Ryan Yates
Marcus Rashford
Justin Kluivert
Lewis Dunk
William Saliba
Mohamed Salah
Virgil van Dijk
João Pedro
Gabriel Magalhães
Danny Welbeck
Edward Nketiah
Darwin Núñez
Diogo Jota
Conor Coady
Raúl Jiménez
Nicolas Jackson
Liam Delap
Nikola Milenković
Michael Keane
Alexander Isak
Jean-Philippe Mateta
Rodrigo Muniz
Jamie Vardy
Dominic Calvert-Lewin
Beto
Ollie Watkins
Erling Haaland
Chris Wood


PCA Projection (PC3 → size/color; exceptional names shown)
PC1
PC2
Trent Alexander-Arnold
Mikkel Damsgaard
Kieran Trippier
Mateo Kovačić
Pedro Porro
Enzo Fernández
Christian Eriksen
Reece James
Elliot Anderson
Mats Wieffer
Manuel Ugarte
İlkay Gündoğan
João Gomes
Will Hughes
Tyler Adams
Nicolás Domínguez
Manuel Akanji
Joško Gvardiol
Rúben Dias
Wilfred Ndidi
Joachim Andersen
Ryan Yates
Lewis Dunk
William Saliba
Virgil van Dijk
Jannik Vestergaard
Tosin Adarabioyo
Gabriel Magalhães
Conor Coady
Nikola Milenković
Michael Keane


PCA Projection (PC3 → size/color; exceptional names shown)
Edward Nketiah
Darwin Núñez
Diogo Jota
Kevin Schade
Raúl Jiménez
Kai Havertz
Dominic Solanke
Nicolas Jackson
Liam Delap
Yoane Wissa
Jean-Philippe Mateta
Jørgen Strand Larsen
Paul Onuachu
Evanilson
Rodrigo Muniz
Jamie Vardy
Dominic Calvert-Lewin
Beto
Ollie Watkins
Erling Haaland
Chris Wood
PC1
PC2
−3
−4
−5
−6
−7
4.0
4.5
5.0
5.5
6.0
6.5
7.0
7.5
8.0


PCA Projection (PC3 → size/color; exceptional names shown)
Jérémy Doku
Kevin De Bruyne
Bruno Fernandes
James Maddison
Bukayo Saka
Savinho
Dwight McNeil
Martin Ødegaard
Facundo Buonanotte
Eberechi Eze
Matheus Cunha
Cole Palmer
Phil Foden
Julio Enciso
Heung-min Son
Jordan Ayew
Luis Díaz
Omar Marmoush
Bryan Mbeumo
Jarrod Bowen
Georginio Rutter
Marcus Rashford
Justin Kluivert
João Pedro
Danny Welbeck
PC1
PC2
6
4
2
0
−2
3
4
5
6
7
8

Example 11 (con't).

- After removing goalkeepers, the first three PCs explain around 45.5% of the total variance. So the plot inevitably misses some information.
- Nevertheless, it highlights most famous top performers, and it also explores some less well-known ones.
- It also appears to distinguish different types of players (defenders, midfielders, forwards) pretty well.

A PCA plot is only a partial view

- A 2D or 3D PCA plot can reveal groups, unusual observations or broad patterns.
- Check ρ_2 or ρ_3: two or three PCs need not retain most of the variance.
- Points that overlap may differ substantially in discarded directions.
- A plot need not preserve all distances or reveal every distinction between classes.

Takeaways

- Covariance records variation after centering.
- Its eigenvectors give orthogonal directions; eigenvalues give their variances.
- PCA retains the leading directions and minimizes average squared reconstruction error.
- Explained variance helps choose k, but does not guarantee predictive usefulness.